

Homework for week 6

1. Exercises from the Kreyszig textbook (we covered some of these in the exercise class)

Chapter 1.1, Problems 1-16
Chapter 1.3, Problems 2-6, 11-15

2. Dropped ball. The differential equation describing a falling body with air resistance reads (with the positive direction pointing downwards)

$$\dot{v}(t) = g - k v(t)^2 .$$

- a) Draw and analyse the direction field of this equation.
b) Verify that the function

$$v(t) = \bar{v} \frac{1 - C e^{-2k\bar{v}t}}{1 + C e^{-2k\bar{v}t}}$$

solves this equation with $\bar{v} = \sqrt{g/k}$ being the terminal velocity.

- c) Show that C is fixed by the initial velocity $v(0) = v_0 > 0$,

$$C = \frac{\bar{v} - v_0}{\bar{v} + v_0} .$$

Bonus questions:

- d) Show that for large times, $t \gg k\bar{v}$, $v(t) \approx \bar{v} (1 - 2Ce^{-2k\bar{v}t})$.
e) Show that for short times, $t \ll k\bar{v}$, $v(t) \approx v_0 + gt - kv_0^2t$.
f) Assume that $v_0 \gg \bar{v}$. Then g can be neglected for short times. Show that the solution of the resulting simpler equation is

$$v(t) = \frac{v_0}{1 + kv_0t} .$$

Note that the short time behaviour of this function agrees with that found in part e) with $g = 0$.